

Revision Notes for Class 8 Maths Chapter 5 Number Play

Number Play is an exciting chapter that explores the patterns and rules around multiples, divisibility, and sums of numbers. In this chapter, students investigate questions like whether all numbers can be written as a sum of consecutive numbers, which numbers can be written in multiple ways as such sums, and how negative numbers might play a role.

- Examples of numbers expressed as sums of consecutive numbers: $7 = 3 + 4$; $10 = 1 + 2 + 3 + 4$; $15 = 1 + 2 + 3 + 4 + 5$.
- Odd numbers can always be written as a sum of two consecutive numbers, however, some even numbers may not always be expressed using only positive consecutive numbers.

Number Play with Pluses and Minuses

When you take any four consecutive numbers, like 3, 4, 5, and 6, and insert '+' and '-' signs in between, you create eight possible expressions. These include combinations like $3 + 4 + 5 + 6$, $3 - 4 - 5 - 6$, and others. Evaluating these reveals something interesting—no matter which set of four consecutive numbers you start with, all resulting sums or differences are even numbers.

Three explanations clarify why this happens:

- Switching a sign flips the value by an even amount (like $2b$), always keeping the parity even.
- Parity rules: $\text{Odd} \pm \text{Odd}$ always yields even, and so does $\text{Even} \pm \text{Even}$.
- Using a token method, all combinations of $a \pm b \pm c \pm d$ lead to even results.

Breaking Even – Which Expressions Are Always Even?

Several algebraic expressions always give an even result for any integer values. For example, $2a + 2b$ and $4m + 2n$ will always be even because each term is a multiple of 2. By writing such expressions in the form $2 \times \text{something}$, it's clear why their total is always even. Expressions like $x^2 + 2$, however, are not always even, since x can be odd or even.

Pairs to Make Fours

If you add together two even numbers, when is their sum divisible by 4? If both numbers are multiples of 4 (like $4p$ and $4q$), their sum is always a multiple of 4. If both are of the form $4p+2$, their sum is also a multiple of 4. But if they are different types (one is $4p$, other is $4q+2$), their sum is not always divisible by 4.

Always, Sometimes, or Never

The chapter includes interesting “Always/Sometimes/Never” statements about divisibility. For example, if 8 divides two numbers, it will always divide their sum. If a number is divisible by 12, it is always divisible by all factors of 12. However, adding an odd and an even never gives a multiple of 6, since their sum is always odd.

What Remains?

To find numbers that leave a remainder of 3 when divided by 5, use the formula $5k + 3$. This sequence gives numbers like 3, 8, 13, 18, and so on. Similarly, numbers that leave particular remainders when divided by different divisors can be found by writing out suitable algebraic expressions, like $5k - 2$ for the same condition starting at higher values of k .

Figure it Out – Problem-Solving Section

Students are encouraged to solve a variety of reasoning questions:

- If the sum of four consecutive numbers is 34, what are the numbers? (Answer: 7, 8, 9, 10).
- If p is the largest of five consecutive numbers, the others are $p-4$, $p-3$, $p-2$, and $p-1$.
- Find numbers with specific remainders (e.g., remainder 2 when divided by 3 and remainder 2 when divided by 4) and write expressions for such numbers.

This section also features questions on divisibility properties, cryptarithms (number puzzles where letters represent digits), and pattern problems around sums and products of consecutive numbers.

Checking Divisibility Quickly

Shortcuts help you determine divisibility fast:

- Divisible by 2: last digit is even.
- Divisible by 5: last digit is 0 or 5.
- Divisible by 10: last digit is 0.
- By 4: last two digits form a number divisible by 4.
- By 8: last three digits form a number divisible by 8.

For 9 or 3, sum the digits: if the result is a multiple of that number, so is the original number! For 11, alternate the sum and subtraction of digits; if the final result is 0 or 11, the number divides by 11.

Divisibility by 6 and 24

A number is divisible by 6 if it is divisible by both 2 and 3. For divisibility by 24, a number must be divisible by both 3 and 8. These combined rules help in quickly checking larger numbers.

Digital Roots

Digital roots involve repeatedly adding the digits of a number until a single-digit result is left. This can help in checking divisibility by 9 – the digital root of a multiple of 9 is always 9.

Cryptarithms – Digits in Disguise

Puzzles where each letter represents a unique digit (and numbers do not start with zero) allow brains to flex! For example, finding digits that solve $A1 + 1B = B0$ or working out what $BYE \times 6 = RAY$ means.

Summary Key Points

- If a divides b , then any multiple of a divides b .
- If a is divisible by both b and c , then a is divisible by $\text{LCM}(b, c)$.

- Sum and differences of numbers divisible by a , remain divisible by a .
- Algebra, counterexamples, and visualisations are powerful tools for thinking about numbers.

Navakankari – A Classic Game

Navakankari (also known as Nine Men's Morris) is an ancient board game where two players aim to form lines of three with their pawns, removing or blocking the opponent's pawns. This game builds logical thinking and strategic skills, tying back to the mathematical mindset encouraged throughout Number Play.